

Introduction

A **band-gap energy** is the **energy difference** between the highest valence band of the material and the lowest conduction band. Conductors have a band-gap of 0 allowing for **free movemnt of electrons** (Sadeghfar & Ghaedi, 2021).

For semiconductors, a **narrower band-gap** allows for that **movement** of electron to be done with **less energy** (Ayyub & Rao, 2020).

Iron-doping, which is the introduction of iron to create instabilities in a material, have been shown to **increase effectiveness** of photocatalysts, however, their **band-gaps** have **not been analyzed independent** of those reaction (Fang et al., 2024).

Purpose

To use semiconductors for more **general applications**, this research is attempting to find the affect of iron-doping on semiconductor band-gap energies. The semiconductors chosen are those most **commonly used** as base **catalysts**.

Research Question

What effect does iron-doping have on the band-gap energies of commonly used photocatalysts?

Hypothesis

Iron-doping will result in a reduction of band-gaps across semiconductors, consistent with performance in photocatalysts.

Variables

IV: Process of Iron-doping
DV: Band-gap energy
Control: Materials without iron-doping

The Effect of Iron-doping on the Band-Gap Energies of TiO2, ZnO, and SnO2

Methods

Experimental Groups:

- 1.Fe-TiO2
- 2.Fe-ZnO
- 3.Fe-SnO2

Control Groups:

- 1.TiO2
- 2.ZnO
- 3.SnO2

- Procedure for synthesis of Fe-TiO2:
- 1. 4.0mL TiCl4 will be added to 400mL of ice water, and will be put under continuous stirring
 - 2.The solution will be stirred for 24 hours after that point.
 - 3.Ammonia Aqueous (1:9) is added afterwards to adjust the pH
 - 4.The precipitation from the sample will then be dleaned with distilled water multiple times
 - 5.30% Hydrogen Peroxide is added until the sample becomes yellow and then transparent

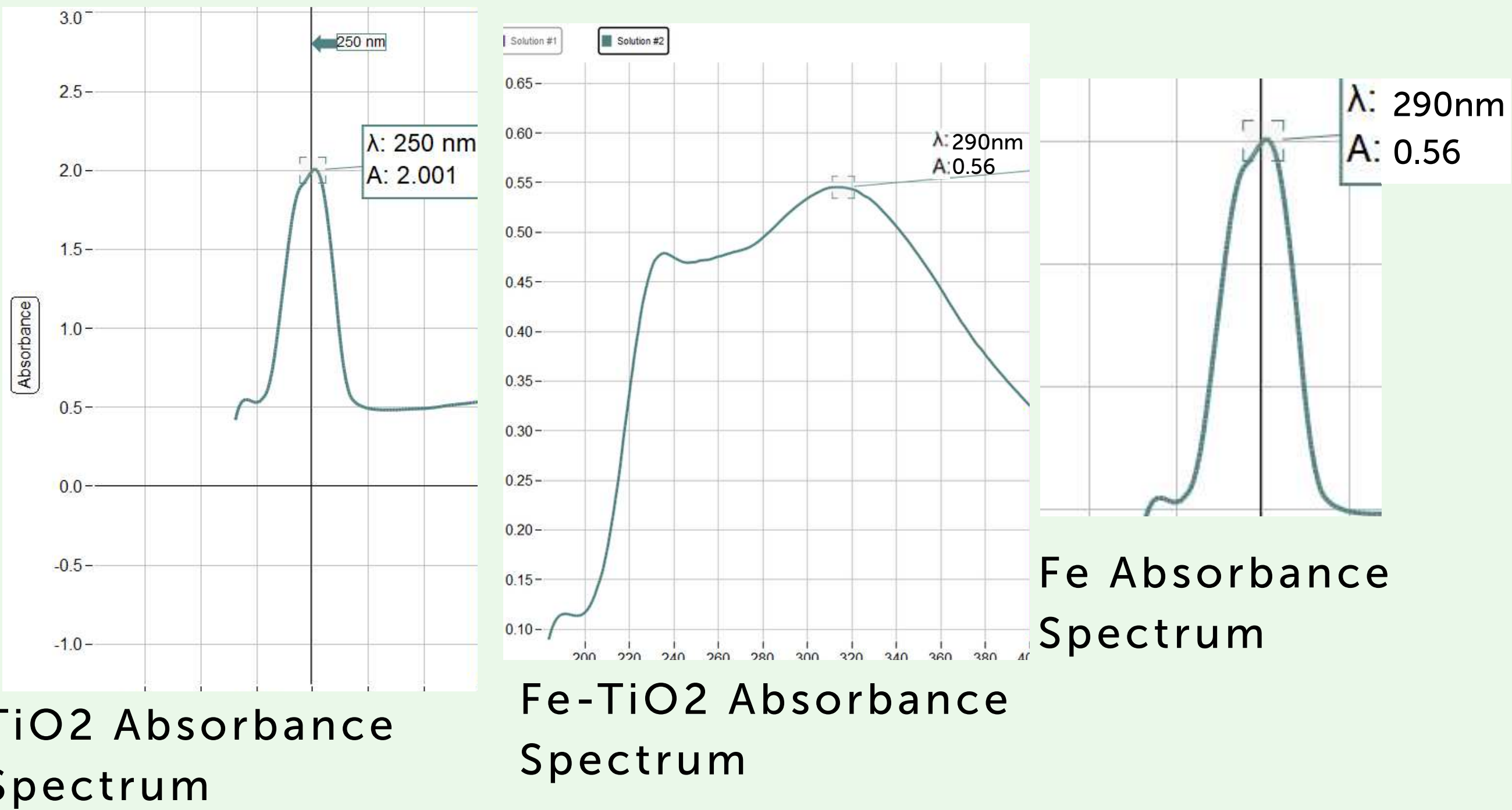
All other chemicals follow similar procedures.

Tauc-plot Analysis Process:

$(ahv)^{1/y} = hv$

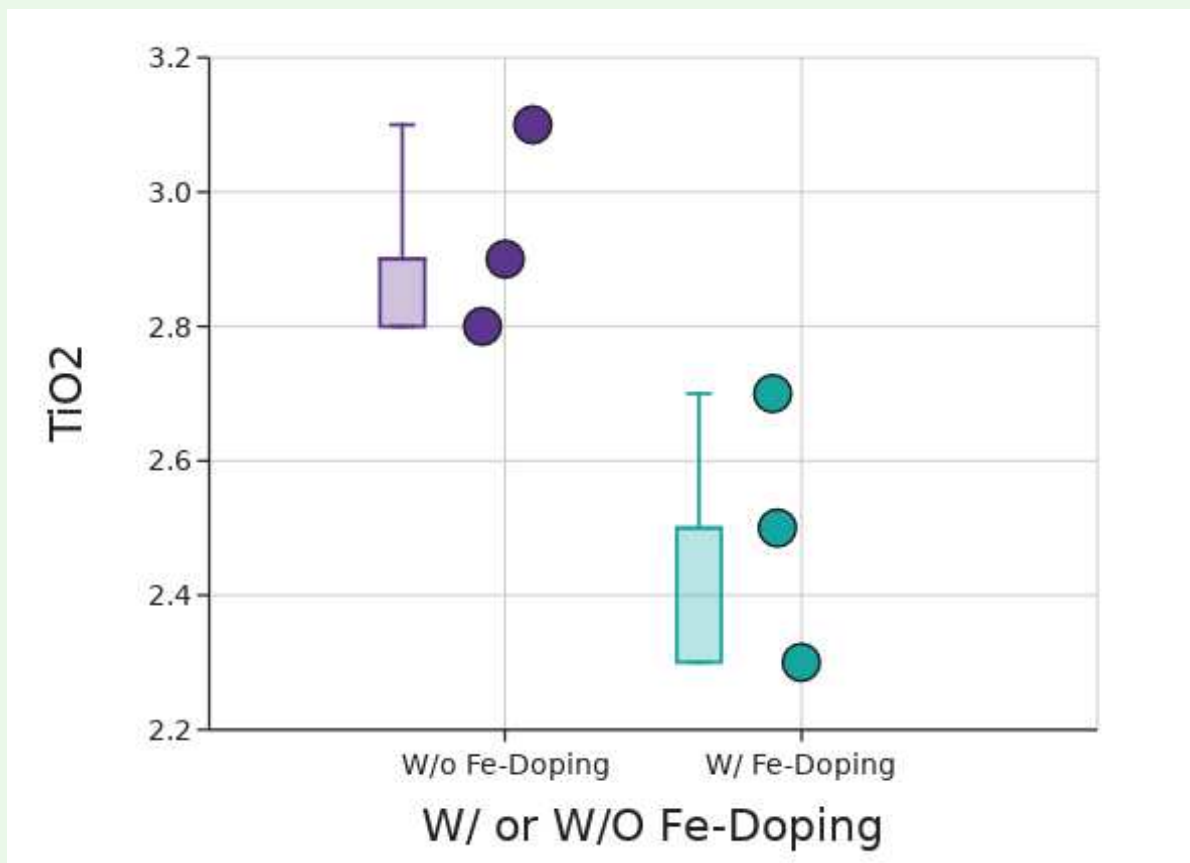
a = Absorption coefficient
y = Transition type

hv = Band-gap energy
hv = 1240/y
y = Wavelength (nm)

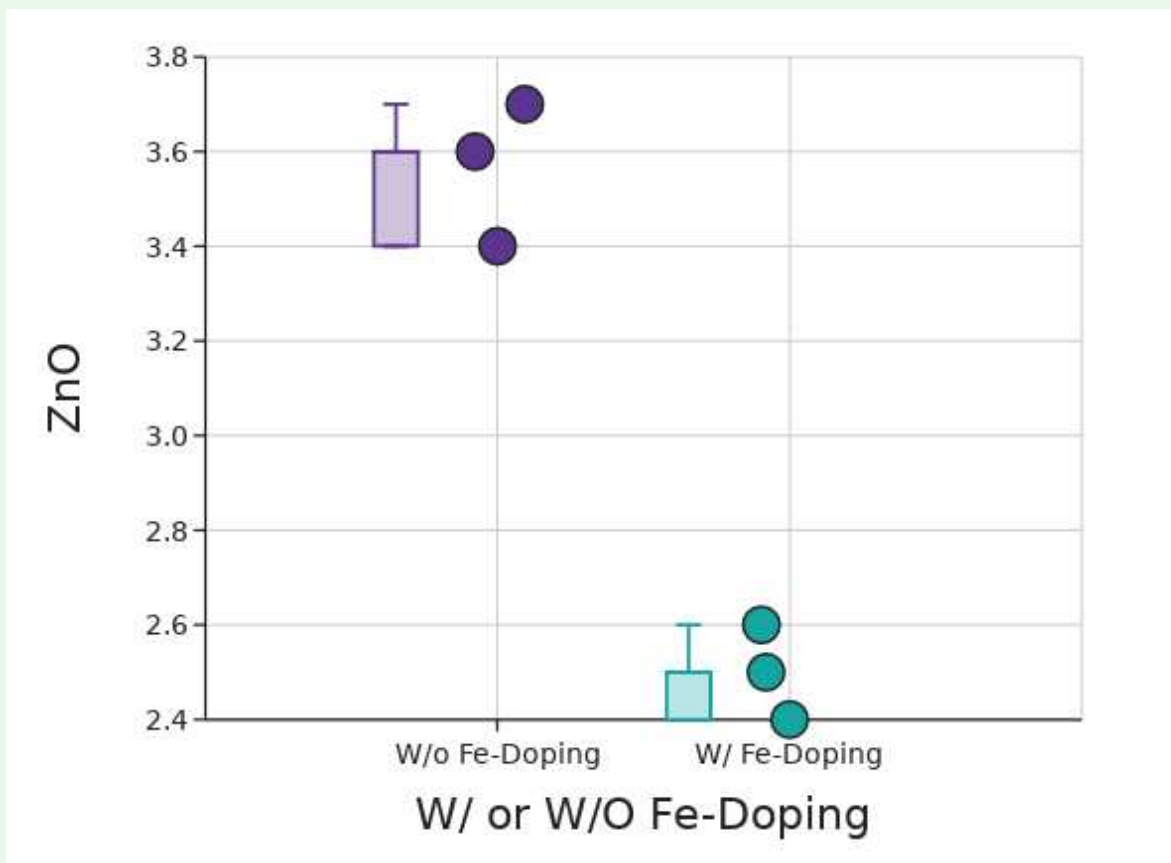


Results

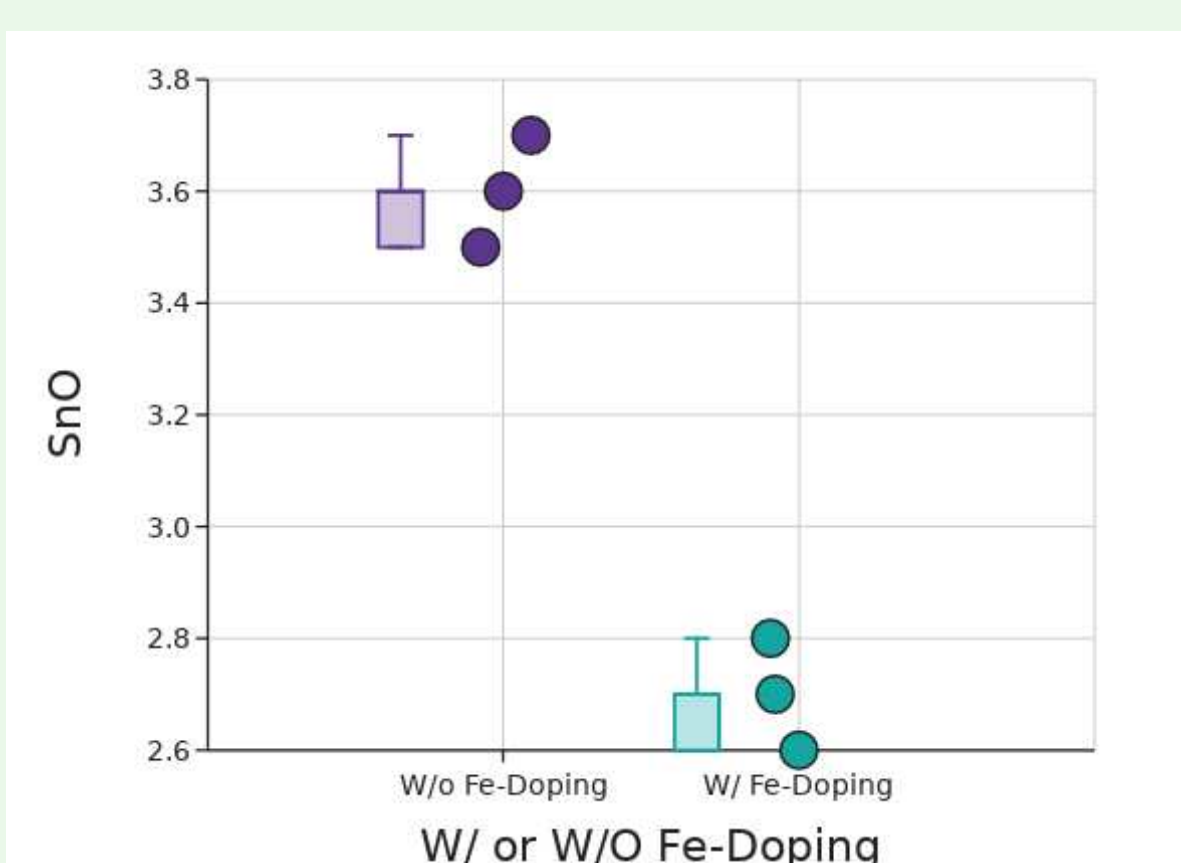
Band-Gap of TiO2 Compared w/ and w/o Iron Doping



Band-Gap of ZnO Compared w/ and w/o Iron Doping



Band-Gap of SnO2 Compared w/ and w/o Iron Doping



Conclusion

Comparing the band-gap energy of each semiconductor--**ZnO, TiO2, and SnO2**--showed significant differences, with **p-values** of **0.03, 0.02, and 0.03** respectively, according to the Mann-Whitney U test.

The actual numerical difference was then plotted, and tested with the Shapiro-Wilk test, which showed a p-value of 0.835, which **failed to reject the null hypothesis that the data is normally distributed**.

This suggests that **iron-doping provides a consistent reduction** to band-gaps, which then raises the band-gap energy to materials manipulated in this way.

For **future** research, testing more semiconductors with a greater number of trials can **test the validity** of a normal distribution of iron-doping capabilities.

If it remains consistent, this could then be **implemented at a larger scale** for use in **batteries, catalysts, manufacturing, and electrical engineering**.

References

Sadeghfar, F., & Ghaedi, M. (2021). Photocatalytic treatment of pollutants in aqueous media. Interface Science and Technology, 725–759. <https://doi.org/10.1016/b978-0-12-818806-4.00011-5>

Ayyub, M. M., & Rao, C. N. R. (2020). Design of efficient photocatalysts through band Gap Engineering. Nanostructured Photocatalysts, 1–18. <https://doi.org/10.1016/b978-0-12-817836-2.00001-6>

Fang, W., Yan, J., Wei, Z., Liu, J., Guo, W., Jiang, Z., & Shangguan, W. (2024). Account of doping photocatalyst for water splitting. Chinese Journal of Catalysis, 60, 1–24. [https://doi.org/10.1016/s1872-2067\(23\)64637-6](https://doi.org/10.1016/s1872-2067(23)64637-6)